

Understanding Bias, the Q-Point, Signal Swing, and Operating Regions in a Common-Emitter Amplifier

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1 The central idea

A transistor amplifier takes a small changing input and produces a larger changing output. To do that faithfully, the transistor must first be placed at a suitable *resting operating point*. An input signal then moves the transistor's current and voltage around that point.

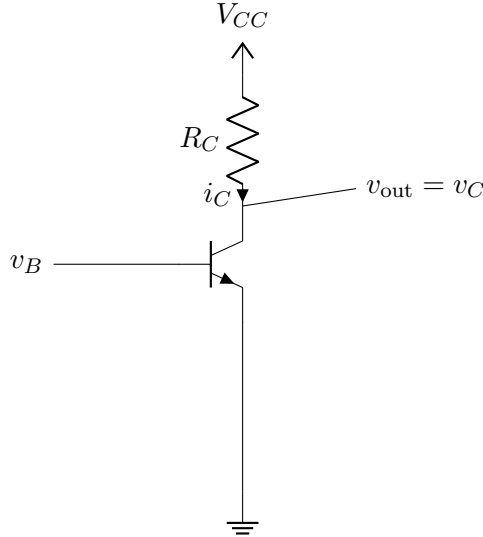
We will distinguish four related ideas:

- **Bias:** The steady voltages and currents applied with no input signal.
- **Q-point:** The resulting resting values of transistor current and voltage. “Q” stands for quiescent.
- **Signal swing:** How far the current or voltage moves above and below its Q-point when a signal is applied.
- **Operating region:** Whether the transistor is in cutoff, forward-active operation, or saturation at a particular instant.

Intuition. Imagine putting a person in the middle of a corridor. The person's starting position is the Q-point. The corridor's walls are cutoff and saturation. The signal asks the person to move back and forth. Starting near the middle permits a larger movement in both directions without hitting a wall.

2 Basic common-emitter circuit without an emitter resistor

Consider the following NPN amplifier. The emitter is grounded, the collector has a resistor R_C , and the output is measured at the collector.



The drawing shows the bias applied at the base as v_B , without committing to a particular bias network. In a practical circuit, a voltage divider and often an emitter resistor provide a more stable bias than a single base resistor.

For now, use

$$V_{CC} = 12 \text{ V}, \quad R_C = 1 \text{ k}\Omega.$$

Because the emitter is at 0 V,

$$V_{CE} = V_C.$$

Applying Kirchhoff's voltage law around the collector circuit gives

$$V_{CC} = I_C R_C + V_{CE}, \quad \text{or} \quad \boxed{V_{CE} = V_{CC} - I_C R_C.} \quad (1)$$

This equation is the circuit's *DC load line*. It tells us that increasing collector current decreases collector voltage.

3 Choosing and interpreting the Q-point

Suppose the bias network sets the resting collector current to

$$I_{CQ} = 6 \text{ mA}.$$

Equation (1) then gives

$$V_{CEQ} = 12 \text{ V} - (6 \text{ mA})(1 \text{ k}\Omega) = 6 \text{ V}.$$

Thus the Q-point is

$$\boxed{(V_{CEQ}, I_{CQ}) = (6 \text{ V}, 6 \text{ mA}).} \quad (2)$$

The subscript Q means “measured with the AC input set to zero.”

For a rough illustration, if the transistor's current gain at this point is $\beta = 100$, its resting base current is approximately

$$I_{BQ} \approx \frac{I_{CQ}}{\beta} = \frac{6 \text{ mA}}{100} = 60 \text{ }\mu\text{A}.$$

A silicon transistor conducting at this current might have V_{BE} around 0.7 V, but 0.7 V is only a convenient approximation: V_{BE} depends on current and temperature.

Why does bias matter? With no bias, a small AC input near zero may leave the transistor off for much of the cycle. Bias establishes a nonzero collector current so that the input can both increase and decrease the current while the transistor remains in its amplifying region.

4 Cutoff, forward-active operation, and saturation

The three regions have different physical meanings.

4.1 Cutoff: essentially off

When the base-emitter junction is not sufficiently forward biased, the collector current is approximately zero:

$$I_C \approx 0.$$

With almost no current through R_C , there is almost no resistor voltage drop. Therefore,

$$V_{CE} \approx V_{CC} = 12 \text{ V}.$$

So the cutoff end of the load line is approximately

$$(V_{CE}, I_C) = (12 \text{ V}, 0).$$

4.2 Forward-active region: useful amplification

In forward-active operation, the base-emitter junction is forward biased and the base-collector junction is reverse biased. Changes in base drive produce changes in collector current. In a simple approximate model,

$$I_C \approx \beta I_B,$$

although β is not perfectly constant and this relation alone does not describe the transistor's detailed response to a voltage input.

For our amplifier, the desired signal should keep the transistor in this region over the full cycle.

4.3 Saturation: driven fully on

If the base drive becomes sufficiently large, the collector voltage falls close to the emitter voltage. Both transistor junctions become forward biased, and the collector current is then limited mainly by the external circuit. Taking $V_{CE(\text{sat})} \approx 0.2 \text{ V}$,

$$I_{C(\text{sat})} \approx \frac{V_{CC} - V_{CE(\text{sat})}}{R_C} = \frac{12 - 0.2}{1 \text{ k}\Omega} = 11.8 \text{ mA}.$$

Thus the saturation end is approximately

$$(V_{CE}, I_C) = (0.2 \text{ V}, 11.8 \text{ mA}).$$

The 0.2 V figure is illustrative; the actual value depends on the transistor and its current.

Region	Meaning	I_C	V_{CE}
Cutoff	Transistor essentially off	≈ 0	$\approx 12\text{ V}$
Forward-active	Collector current responds to input	Between limits	Between limits
Q-point	Resting point within forward-active region	6 mA	6 V
Saturation	Transistor driven strongly on	$\approx 11.8\text{ mA}$	$\approx 0.2\text{ V}$

4.4 Why do we use $V_{CE(\text{sat})} \approx 0.2\text{ V}$?

The value 0.2 V is an *approximation for a silicon BJT in saturation*, not a voltage that every transistor always has.

In forward-active operation, the base-emitter junction is forward biased while the base-collector junction is reverse biased. As base drive increases, collector current increases and the collector voltage falls. Eventually the collector voltage becomes low enough that the base-collector junction also becomes forward biased. The transistor has entered *saturation*.

The transistor voltages obey

$$V_{CE} = V_C - V_E = (V_B - V_E) - (V_B - V_C) = V_{BE} - V_{BC}.$$

In saturation, both V_{BE} and V_{BC} are positive because both junctions are forward biased. Their difference is usually small. For example, illustrative values

$$V_{BE} = 0.8\text{ V}, \quad V_{BC} = 0.6\text{ V}$$

give

$$V_{CE} = 0.8 - 0.6 = 0.2\text{ V}.$$

This is the intuition behind the commonly used estimate

$$\boxed{V_{CE(\text{sat})} \approx 0.2\text{ V}.$$

The actual saturation voltage depends on the transistor, collector current, temperature, and amount of base drive. Its datasheet may specify $V_{CE(\text{sat})}$ under particular test conditions.

What saturation means for our circuit. Once the transistor is saturated, increasing base drive cannot make the collector current continue to follow $I_C \approx \beta I_B$. The supply and external resistors largely limit the current instead. For the grounded-emitter example,

$$I_{C(\text{sat})} \approx \frac{V_{CC} - V_{CE(\text{sat})}}{R_C} = \frac{12 - 0.2}{1\text{ k}\Omega} = 11.8\text{ mA}.$$

The collector voltage cannot fall much further, so an amplifier signal driven beyond this point clips.

For the example with an emitter resistor, the emitter also rises above ground. Using $I_E \approx I_C$, the corresponding estimate is

$$I_{C(\text{sat})} \approx \frac{V_{CC} - V_{CE(\text{sat})}}{R_C + R_E} = \frac{12 - 0.2}{1.5\text{ k}\Omega} \approx 7.87\text{ mA}.$$

In the follow-up small-signal document, we model the transistor *near its forward-active Q-point*. That linear model should not be extended into saturation: the 0.2 V estimate instead helps us identify where the predicted signal swing must stop.

5 What is signal swing?

The input signal makes current vary around I_{CQ} . For an illustrative sinusoidal collector current, write

$$i_C(t) = I_{CQ} + I_m \sin(\omega t), \quad (3)$$

where I_m is the *peak AC current amplitude*, not the total current.

Using the collector resistor relation,

$$v_{CE}(t) = V_{CC} - i_C(t)R_C \quad (4)$$

$$= \underbrace{(V_{CC} - I_{CQ}R_C)}_{V_{CEQ}} - I_m R_C \sin(\omega t). \quad (5)$$

Therefore,

$$v_{CE}(t) = V_{CEQ} - V_m \sin(\omega t), \quad V_m = I_m R_C. \quad (6)$$

The current and collector voltage move in opposite directions. This is the familiar **phase inversion** of a common-emitter stage.

Numerical example: a moderate signal

Suppose $I_m = 2$ mA. Then $V_m = (2 \text{ mA})(1 \text{ k}\Omega) = 2$ V, giving

$$i_C(t) = 6 \text{ mA} + 2 \text{ mA} \sin(\omega t),$$

$$v_{CE}(t) = 6 \text{ V} - 2 \text{ V} \sin(\omega t).$$

Instant	Collector current	Collector voltage
Resting value	6 mA	6 V
Maximum current	8 mA	4 V
Minimum current	4 mA	8 V

Both extremes remain away from cutoff and saturation. The *peak-to-peak* current swing is 4 mA, while the peak-to-peak collector-voltage swing is 4 V.

Remember the distinction: $I_{CQ} = 6$ mA is the steady current; $I_m = 2$ mA is the signal's peak amplitude. The instantaneous current moves from $I_{CQ} - I_m$ to $I_{CQ} + I_m$.

6 Maximum swing and clipping

From the chosen Q-point, the current can fall from 6 mA to approximately zero before cutoff. Thus the cutoff side permits a peak current change of 6 mA.

On the other side, current can rise from 6 mA to approximately 11.8 mA before saturation. That side permits a peak change of 5.8 mA.

Consequently, with the Q-point chosen above, the largest approximately symmetric swing allowed by these simple limits is

$$I_m \approx 5.8 \text{ mA}, \quad V_m = I_m R_C \approx 5.8 \text{ V}.$$

The output would then vary approximately between 0.2 V and 11.8 V. It is common in introductory calculations to neglect $V_{CE(\text{sat})}$ and say $I_m \approx I_{CQ}$ and $V_m \approx V_{CEQ}$. The distinction between 5.8 and 6 here comes from retaining the illustrative 0.2 V saturation voltage.

If a larger input asks for $I_C > 11.8$ mA, the collector voltage cannot fall much below its saturation value. If the input asks for $I_C < 0$, the transistor instead enters cutoff. Either case causes *clipping*: part of the output waveform becomes flattened.

Where exactly is the midpoint?

With cutoff at $V_{CE} = 12$ V and saturation at $V_{CE} = 0.2$ V, the voltage midpoint is

$$V_{CEQ,\text{mid}} = \frac{12 + 0.2}{2} = 6.1 \text{ V.}$$

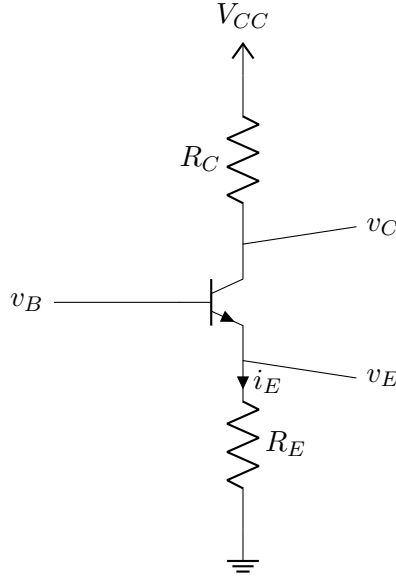
The corresponding current is

$$I_{CQ,\text{mid}} = \frac{12 - 6.1}{1 \text{ k}\Omega} = 5.9 \text{ mA.}$$

Thus (6.1 V, 5.9 mA) gives exactly equal room in the two directions under this simplified load-line model. The earlier (6 V, 6 mA) choice is a convenient near-midpoint example.

7 Special case: adding an emitter resistor

Now connect a resistor R_E between the emitter and ground. The emitter is no longer necessarily at 0 V.



There are now *three* important voltages:

$$V_C = V_{CC} - I_C R_C, \quad V_E = I_E R_E, \quad \boxed{V_{CE} = V_C - V_E.}$$

The base must sit above the emitter by approximately V_{BE} , so

$$V_B = V_E + V_{BE} \approx I_E R_E + 0.7 \text{ V.}$$

Since $I_E = I_C + I_B$, for a transistor with reasonably large β , we often use $I_E \approx I_C$. This gives the useful approximate DC relation

$$\boxed{V_{CE} \approx V_{CC} - I_C (R_C + R_E).} \quad (7)$$

Equation (7) is for the *DC circuit*, where both resistors carry approximately the same transistor current.

7.1 A complete numerical example with R_E

Let

$$V_{CC} = 12 \text{ V}, \quad R_C = 1 \text{ k}\Omega, \quad R_E = 500 \text{ }\Omega.$$

Suppose the bias circuit establishes $I_{CQ} \approx 4 \text{ mA}$. With $I_E \approx I_C$,

$$\begin{aligned} V_{EQ} &\approx (4 \text{ mA})(500 \text{ }\Omega) = 2 \text{ V}, \\ V_{CQ} &= 12 \text{ V} - (4 \text{ mA})(1 \text{ k}\Omega) = 8 \text{ V}, \\ V_{CEQ} &= V_{CQ} - V_{EQ} = 8 - 2 = 6 \text{ V}, \\ V_{BQ} &\approx V_{EQ} + 0.7 \text{ V} = 2.7 \text{ V}. \end{aligned}$$

The Q-point in the transistor's output characteristics is therefore

$$(V_{CEQ}, I_{CQ}) \approx (6 \text{ V}, 4 \text{ mA}).$$

Notice that $V_C = 8 \text{ V}$, whereas $V_{CE} = 6 \text{ V}$. These were equal in the grounded-emitter example, but they are *different* when an emitter resistor raises the emitter to 2 V.

7.2 Where are cutoff and saturation now?

At cutoff, $I_C \approx I_E \approx 0$. Consequently $V_E \approx 0$, $V_C \approx 12 \text{ V}$, and

$$V_{CE} \approx 12 \text{ V}.$$

Near saturation, taking $V_{CE(\text{sat})} \approx 0.2 \text{ V}$ and $I_E \approx I_C$, the DC current is approximately

$$I_{C(\text{sat})} \approx \frac{V_{CC} - V_{CE(\text{sat})}}{R_C + R_E} = \frac{11.8 \text{ V}}{1.5 \text{ k}\Omega} \approx 7.87 \text{ mA}.$$

At that current, $V_E \approx 3.93 \text{ V}$ and $V_C \approx 4.13 \text{ V}$, leaving approximately 0.2 V between collector and emitter.

From $I_{CQ} = 4 \text{ mA}$, the current has room to decrease by 4 mA before cutoff and increase by approximately 3.87 mA before saturation. The chosen Q-point is again near the middle of the available range.

7.3 Why does an emitter resistor stabilize the bias?

Suppose temperature or transistor variation tries to make I_C increase. Then I_E also increases, causing a larger drop across R_E :

$$V_E = I_E R_E \text{ increases.}$$

If the bias circuit keeps V_B approximately fixed, then

$$V_{BE} = V_B - V_E \text{ decreases.}$$

That reduction in V_{BE} opposes the original current increase. This is *negative feedback*. For an especially simple bias estimate, if the base is held near $V_B = 2.7 \text{ V}$, then

$$I_E \approx \frac{V_B - 0.7 \text{ V}}{R_E} = \frac{2.0 \text{ V}}{500 \text{ }\Omega} = 4 \text{ mA}.$$

The current is now set largely by a voltage and a resistor, making it less dependent on the uncertain value of β . A real voltage-divider bias must also be designed to supply the base current without excessive loading.

7.4 What happens to AC gain?

If R_E is *not bypassed* by a capacitor, the AC signal also produces a changing emitter voltage. When base voltage rises, collector and emitter current rise, which raises v_E . This reduces the change in $v_{BE} = v_B - v_E$. The feedback therefore reduces gain but improves stability and linearity.

For a moderate-frequency small-signal approximation, assuming the collector has no additional AC load, the voltage gain from base to collector is roughly

$$A_v \approx -\frac{R_C}{r_e + R_E}, \quad r_e \approx \frac{25-26 \text{ mV}}{I_{EQ}} \quad \text{at room temperature.} \quad (8)$$

At $I_{EQ} \approx 4 \text{ mA}$, r_e is about 6.5Ω , so this rough estimate gives

$$A_v \approx -\frac{1000}{500 + 6.5} \approx -2.0.$$

The minus sign again indicates inversion. This approximate formula assumes a sufficiently large β , a suitably stiff base drive, and no significant output loading.

If a capacitor effectively *bypasses* R_E for the AC signal, the emitter resistor still helps establish the DC Q-point, while much of its AC feedback is removed. The gain can then be much larger. The capacitor's effectiveness depends on signal frequency.

Important distinction: The DC load line includes R_E . For an AC signal, whether R_E affects the signal swing depends on the bypass capacitor and the rest of the circuit. Do not automatically use the DC load-line slope as the AC load-line slope.

8 DC and AC load lines: what each one tells us

A *load line* shows the combinations of collector current I_C and collector-emitter voltage V_{CE} that the external circuit permits. The transistor's characteristics determine where on that line it operates.

There are two related questions:

- **DC load line:** Where can the transistor's resting Q-point be placed?
- **AC load line:** Starting from that Q-point, how far can the signal move in either direction before clipping?

We plot V_{CE} on the horizontal axis and I_C on the vertical axis. A load line slopes downward: greater collector current generally leaves less voltage across the transistor.

8.1 DC load line for the grounded-emitter circuit

For the first example, $V_{CC} = 12 \text{ V}$ and $R_C = 1 \text{ k}\Omega$. Applying Kirchhoff's voltage law gives

$$V_{CE} = V_{CC} - I_C R_C = 12 \text{ V} - I_C(1 \text{ k}\Omega).$$

Two points determine this straight line:

$$\begin{aligned} I_C = 0 &\implies V_{CE} = 12 \text{ V} && \text{(cutoff end),} \\ V_{CE} \approx 0 &\implies I_C \approx 12 \text{ mA} && \text{(idealized saturation end).} \end{aligned}$$

If we retain $V_{CE(\text{sat})} \approx 0.2 \text{ V}$, the practical saturation-end estimate becomes 11.8 mA. Bias establishes the previously chosen Q-point

$$Q : (V_{CEQ}, I_{CQ}) = (6 \text{ V}, 6 \text{ mA})$$

on this DC load line.

Significance of the DC load line. It shows the complete range of possible *steady* operating points for the given supply and resistors. The bias network selects one point from that range. It also shows whether a proposed Q-point is near cutoff, near saturation, or roughly between them.

8.2 Why can the AC load line be different?

Suppose a load resistor R_L is connected to the collector through an output coupling capacitor. For DC, the capacitor is an open circuit: no steady current flows through R_L . Hence R_L does not change the DC load line or, ideally, the Q-point.

At a signal frequency where the capacitor acts approximately as a short circuit, R_L *does* carry AC current. Also, the DC supply V_{CC} is an AC ground: its voltage does not change with the signal. From the collector's AC point of view, R_C and R_L both connect to AC ground and therefore appear in parallel:

$$R_{ac} = R_C \parallel R_L = \frac{R_C R_L}{R_C + R_L}.$$

The signal's changes in voltage and current consequently obey

$$\boxed{\Delta v_{CE} = -(R_C \parallel R_L) \Delta i_C} \quad \text{(grounded emitter).} \quad (9)$$

Because changes are measured relative to the Q-point,

$$\Delta v_{CE} = v_{CE} - V_{CEQ}, \quad \Delta i_C = i_C - I_{CQ}.$$

Thus the AC load line is

$$\boxed{v_{CE} - V_{CEQ} = -(R_C \parallel R_L)(i_C - I_{CQ})}. \quad (10)$$

Both load lines pass through Q, but they need not have the same slope. The DC line determines the resting point; the AC line describes signal movement away from it.

8.3 Numerical example: adding an AC-coupled load

Keep

$$V_{CC} = 12 \text{ V}, \quad R_C = 1 \text{ k}\Omega, \quad Q = (6 \text{ V}, 6 \text{ mA}),$$

and connect $R_L = 1 \text{ k}\Omega$ through a coupling capacitor. Then

$$R_C \parallel R_L = 1 \text{ k}\Omega \parallel 1 \text{ k}\Omega = 500 \text{ }\Omega.$$

The DC load line remains

$$V_{CE} = 12 \text{ V} - I_C(1 \text{ k}\Omega),$$

but the AC load line through Q becomes

$$v_{CE} - 6 \text{ V} = -(500 \Omega)(i_C - 6 \text{ mA}).$$

For example, if the signal raises collector current by 2 mA, the collector voltage falls by only

$$(2 \text{ mA})(500 \Omega) = 1 \text{ V}.$$

The instantaneous operating point moves from (6 V, 6 mA) to (5 V, 8 mA). Without R_L , the same 2 mA current change would produce a 2 V collector-voltage change.

At the cutoff end of the *idealized AC line*, setting $i_C = 0$ gives

$$v_{CE} = 6 \text{ V} + (500 \Omega)(6 \text{ mA}) = 9 \text{ V}.$$

At the saturation boundary, setting $v_{CE} \approx 0.2 \text{ V}$ gives

$$i_C \approx 6 \text{ mA} + \frac{6 - 0.2}{500 \Omega} = 17.6 \text{ mA}.$$

This 17.6 mA is an *instantaneous signal* limit predicted by the AC load line; it is not the DC saturation current of 11.8 mA. The coupling capacitor can supply or absorb signal charge, so the instantaneous current relationships differ from the steady DC relationships. These estimates assume a sufficiently large coupling capacitor and an ideal supply at AC ground.

The available excursions from Q are

$$\text{Toward cutoff: } \Delta I_C = -6 \text{ mA}, \quad \Delta V_{CE} = +3 \text{ V},$$

$$\text{Toward saturation: } \Delta V_{CE} = -5.8 \text{ V}, \quad \Delta I_C = +11.6 \text{ mA}.$$

For a *symmetric sinusoidal current* swing, cutoff therefore limits the peak amplitude to approximately $I_m = 6 \text{ mA}$. The corresponding collector-voltage peak amplitude is only

$$V_m = I_m(R_C \parallel R_L) = (6 \text{ mA})(500 \Omega) = 3 \text{ V}.$$

Practical significance. A Q-point that appears centered on the DC load line is not necessarily centered on the AC load line. Connecting a load can change which half-cycle clips first and can reduce the available output-voltage swing.

8.4 DC load line with an emitter resistor

Return to the emitter-resistor example:

$$V_{CC} = 12 \text{ V}, \quad R_C = 1 \text{ k}\Omega, \quad R_E = 500 \Omega.$$

For DC, Kirchhoff's voltage law gives the exact relation

$$V_{CC} = I_C R_C + V_{CE} + I_E R_E.$$

With the approximation $I_E \approx I_C$,

$$V_{CE} \approx V_{CC} - I_C(R_C + R_E).$$

Its idealized endpoints are

$$I_C = 0 \implies V_{CE} = 12 \text{ V},$$

$$V_{CE} \approx 0 \implies I_C \approx \frac{12 \text{ V}}{1.5 \text{ k}\Omega} = 8 \text{ mA}.$$

The chosen Q-point, $(V_{CEQ}, I_{CQ}) \approx (6 \text{ V}, 4 \text{ mA})$, lies at the midpoint of this idealized DC line.

8.5 AC load line with an unbypassed emitter resistor

Suppose R_E has *no bypass capacitor*. For AC, an increase in collector current raises the emitter voltage as well as lowering the collector voltage. Assuming $i_E \approx i_C$, and with an AC-coupled collector load R_L ,

$$\Delta v_C = -(R_C \parallel R_L) \Delta i_C, \quad \Delta v_E \approx R_E \Delta i_C.$$

Since $v_{CE} = v_C - v_E$,

$$\Delta v_{CE} \approx -[(R_C \parallel R_L) + R_E] \Delta i_C. \quad (11)$$

For the numerical emitter-resistor example, if there is no separate collector load, $R_C \parallel R_L$ is simply R_C . The AC slope is then approximately $-(R_C + R_E) = -1.5 \text{ k}\Omega$, the same as the approximate DC slope.

If instead $R_L = 1 \text{ k}\Omega$, then

$$(R_C \parallel R_L) + R_E = 500 \text{ }\Omega + 500 \text{ }\Omega = 1 \text{ k}\Omega.$$

The AC line through $Q = (6 \text{ V}, 4 \text{ mA})$ is

$$v_{CE} - 6 \text{ V} \approx -(1 \text{ k}\Omega)(i_C - 4 \text{ mA}).$$

For the *collector output voltage* v_C , however, the AC change is only

$$\Delta v_C = -(500 \text{ }\Omega) \Delta i_C.$$

This distinction matters: the AC load line is plotted using $v_{CE} = v_C - v_E$, while the output terminal in our circuit is at the collector.

8.6 What if the emitter resistor is bypassed?

An emitter-bypass capacitor is open for DC, so it does not change the DC Q-point or DC load line. At a frequency where it effectively shorts R_E , the emitter becomes approximately AC-grounded. The AC relation then approaches

$$\Delta v_{CE} \approx -(R_C \parallel R_L) \Delta i_C.$$

Thus bypassing R_E changes the AC line and gain while preserving the DC bias set by R_E . At frequencies where the bypass capacitor is not an effective short, the actual AC behavior lies between the fully bypassed and unbypassed cases.

Summary: which line should I use?

Question	Use	Reason
Where is the Q-point?	DC load line and bias circuit	Capacitors block steady current.
How far can a signal swing?	AC load line through the Q-point	Coupling and bypass capacitors can change the effective AC load.
Where does clipping begin?	AC line and transistor cutoff/saturation boundaries	The signal follows the AC line until it reaches an operating limit.
What happens if R_L changes?	Recalculate the AC load	An AC-coupled R_L may change swing without changing the DC Q-point.

9 Connection to Class A operation

A CE stage biased in the forward-active region can conduct throughout the full signal cycle: this is Class A operation. In the first example, the collector current never becomes negative. Even at the limiting swing, it only just approaches zero at one instant.

The frequently used statement

$$I_m = I_{CQ}$$

describes the *ideal cutoff-side limit* of the current swing. It is not true for an arbitrary input signal. With our moderate-signal example, $I_{CQ} = 6$ mA but $I_m = 2$ mA. Accounting for saturation may also make the allowed symmetric swing slightly less than I_{CQ} .

10 Quick comparison

Quantity	Grounded emitter	Emitter resistor present
Emitter DC voltage	$V_E = 0$	$V_E = I_E R_E$
Collector voltage	$V_C = V_{CC} - I_C R_C$	$V_C = V_{CC} - I_C R_C$
Collector-emitter voltage	$V_{CE} = V_C$	$V_{CE} = V_C - V_E$
Approximate DC load line	$V_{CE} = V_{CC} - I_C R_C$	$V_{CE} \approx V_{CC} - I_C (R_C + R_E)$
Response to current drift	Little local feedback	R_E provides negative feedback
AC gain	Usually larger	Lower if R_E is unbypassed

11 Check your understanding

1. In the first circuit, if $I_C = 9$ mA, what is V_{CE} ? Is this larger or smaller than V_{CEQ} ?
2. If the first circuit is biased at $I_{CQ} = 2$ mA, which side of the possible current swing reaches its limit first?
3. In the emitter-resistor example, why is $V_C = 8$ V while $V_{CE} = 6$ V?
4. If V_B is held approximately fixed and I_C starts increasing, explain how R_E opposes that increase.

Answers. (1) $V_{CE} = 12 - 9 = 3$ V, smaller than the 6 V Q-point value. (2) The current can fall only 2 mA before cutoff, so the cutoff side limits the symmetric swing. (3) The emitter is at 2 V, and $V_{CE} = V_C - V_E = 8 - 2 = 6$ V. (4) Increased current raises V_E , reducing V_{BE} and opposing the increase.